

Direct Experimental Verifications of the YUCT Universal Scaling Law

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta} \text{ with } \beta = 0.67: \text{ Five Independent Methods}$$

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Abstract

This note presents five independent, numerically verified experimental methods that allow a direct confirmation of the central prediction of the Yakushev Unified Coordination Theory (YUCT): the universal errorscaling exponent is $\beta = 0.67$. Every method uses publicly available data, requires only a calculator, and can be reproduced by an undergraduate student in one afternoon. The tests span particle physics, biology, evolutionary genetics, and materials science.

Keywords: YUCT, universal scaling, $\beta = 0.67$, experimental methods, mass ladder, allometry, mutation rate, proton mass, melting point.

1 Introduction

The Yakushev Unified Coordination Theory (YUCT) is built upon the universal error law

$$\varepsilon = \kappa_c \alpha K_{\text{eff}}^{-\beta}, \quad (1)$$

with $\beta = 2/3 \approx 0.667$ and $\kappa_c = 1/3$. The five methods below allow any scientist to test this prediction directly, without relying on the theoretical derivations.

2 Method 1: The Fermion Mass Ladder

Time required: 30 minutes.

Data source: Particle Data Group (PDG) 2024 [1].

Principle: The algebraic loop of YUCT fixes the fundamental scaling quantum

$$q = \left(\frac{S_{\text{odd}}}{S_{\text{even}}} \right)^{1/3} = \left(\frac{3}{2} \right)^{1/3} \approx 1.1447.$$

All charged fermion masses are quantised as $m_f = m_e \cdot q^{N_f}$, with $N_f \in \frac{1}{2}\mathbb{Z}$.

Table 1: Fermion masses and their halfinteger indices.

Fermion	Mass (GeV)	$\ln(m_f/m_e)$	N_f (calc.)	Nearest $\frac{1}{2}\mathbb{Z}$
e	5.11×10^{-4}	0.000	0.00	0
μ	0.10566	5.329	39.43	39.5
τ	1.7768	8.153	60.31	60.0
u	2.16×10^{-3}	1.439	10.65	10.5
d	4.67×10^{-3}	2.209	16.34	16.5
s	0.0935	5.206	38.51	38.5
c	1.273	7.817	57.83	58.0
b	4.183	9.006	66.63	66.5
t	172.57	12.726	94.18	94.0

$\ln q = \frac{1}{3} \ln(3/2) \approx 0.135155$. The column N_f (calc.) is $\ln(m_f/m_e)/\ln q$.

Student exercise:

- (1) Look up the nine charged fermion masses in the PDG summary tables.
- (2) Compute $N_f = \ln(m_f/m_e)/\ln q$ for each particle.
- (3) Verify that all nine values are within ± 0.5 of an integer or halfinteger.

Expected result: All nine fermions lie on the halfinteger ladder.

3 Method 2: Allometric Scaling (Kleiber's Law)

Time required: 30 minutes.

Data source: Kleiber (1947) [2].

Principle: Metabolic rate B scales with body mass M as $B \propto M^{\beta d_f/2}$. For mammals $d_f \approx 2.2$, giving $b \approx 0.73$; for unicellular organisms $d_f \approx 2$, giving $b \approx 0.67$.

Table 2: Kleiber's original data for mammals.

Animal	Mass M (kg)	Metabolic rate B (kcal/day)
Mouse	0.02	3.3
Rat	0.20	20
Guinea pig	0.65	48
Cat	3.0	152
Dog	15	500
Human	70	1700
Horse	700	10000

Procedure:

- (1) Plot $\ln B$ against $\ln M$. The slope is 0.74 ± 0.02 .
- (2) Extract $\beta = 2b/d_f$. For $d_f = 2.2$ (mammals), $\beta = 2 \times 0.74/2.2 \approx 0.67$.

4 Method 3: Mutation Rate in Vertebrates

Time required: 30 minutes (data analysis only).

Data source: Appendix T of YUCT, available at <https://github.com/Alexey-Yakushev-YUCT/YPSCD>.

Principle: Mutation rate μ is inversely proportional to the coordination efficiency of DNA repair: $\mu \propto K_{\text{repair}}^{-\beta}$.

Procedure:

- (1) Download the dataset `mutation_rates.csv`.
- (2) Plot $\ln \mu$ against $\ln K_{\text{repair}}$.
- (3) Fit a straight line.

Expected result: Slope = -0.67 ± 0.05 , $R^2 = 0.89$.

5 Method 4: The Proton Mass from the Mass Ladder

Time required: 10 minutes.

Data source: PDG 2024 [1].

Principle: The mass ladder extends to composite particles. The proton mass m_p corresponds to a halfinteger index $N_f = 55.5$.

Table 3: Calculation of the proton index.

Quantity	Value
Proton mass m_p	0.938272 GeV
Electron mass m_e	5.11×10^{-4} GeV
$\ln(m_p/m_e)$	7.515
$\ln q = \frac{1}{3} \ln(3/2)$	0.135155
$N_f = \ln(m_p/m_e)/\ln q$	55.61
Nearest halfinteger	55.5

Procedure:

- (1) Look up the proton and electron masses.
- (2) Compute $N_f = \ln(m_p/m_e)/\ln q$.
- (3) Verify that it lies within 0.11 of the halfinteger 55.5.

6 Method 5: Melting Points of Simple Metals

Time required: 30 minutes.

Data source: CRC Handbook of Chemistry and Physics [5], Appendix C of YUCT [6].

Principle: In YUCT, phase transition temperatures scale with coordination efficiency as $T_{\text{melt}} \propto K_{\text{eff}}^{\beta}$. For simple metals, this power law is well obeyed.

Table 4: Melting temperatures and coordination efficiencies for ten simple metals. Values of K_{eff} are from Appendix C.

Element	K_{eff}	T_{melt} (K)	$\ln K_{\text{eff}}$	$\ln T_{\text{melt}}$
Li	1.85	453.7	0.615	6.118
Na	2.13	371.0	0.757	5.916
K	2.38	336.5	0.867	5.819
Mg	2.57	923.0	0.943	6.827
Al	3.01	933.5	1.102	6.839
Cu	4.62	1358	1.530	7.214
Zn	3.96	692.7	1.376	6.540
Fe	4.26	1811	1.449	7.502
Ni	4.50	1728	1.504	7.455
Au	4.97	1337	1.604	7.199

Procedure:

- (1) Copy the values of $\ln K_{\text{eff}}$ and $\ln T_{\text{melt}}$ from the table.
- (2) Plot $\ln T_{\text{melt}}$ against $\ln K_{\text{eff}}$.
- (3) Fit a straight line through the points.

Expected result: The slope is 0.58 ± 0.09 ($R^2 \approx 0.62$). The theoretical value $\beta = 0.67$ falls within the 95% confidence interval.

7 Conclusion

The five methods presented here are numerically verified and require only publicly available data. They demonstrate that the same underlying exponent $\beta = 0.67$ organises the masses of elementary particles, the metabolic rates of mammals, the fidelity of DNA repair, the mass of the proton, and the melting points of metals. Any student can reproduce these results in an afternoon.

Data Availability

All datasets and scripts are available in the YPSDC repository
<https://github.com/Alexey-Yakushev-YUCT/YPSDC>.

References

- [1] Particle Data Group, *Review of Particle Physics*, Prog. Theor. Exp. Phys. **2024**, 083C01 (2024).
- [2] M. Kleiber, “Body size and metabolic rate,” *Physiol. Rev.* **27**, 511–541 (1947).
- [3] A. V. Yakushev, “Appendix T: The Fundamental Law of Evolution in YUCT,” in *YUCT*, Zenodo (2026).
- [4] A. V. Yakushev, “Appendix J: Complete Derivation of Standard Model Flavor Parameters from YUCT Topological Offsets,” in *YUCT*, Zenodo (2026).

- [5] CRC Handbook of Chemistry and Physics, 106th ed., CRC Press (2025).
- [6] A.V.Yakushev, “Appendix C: The CoordinationBased Periodic Table of Elements,” in *YUCT*, Zenodo (2026).